Computer Organization and Architecture

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Function Parameters

Which register is used for the function parameter `val1` in:

```
float func(double val1, double* val2, float val3,
float* val4, int val5, double val6)
```

- A `x0`
- B `w0`
- C `s0`
- D `d0`
- E `d1`

Function Parameters

Which register is used for the function parameter `val2` in:

```
float func(double val1, double* val2, float val3,
float* val4, int val5, double val6)
```

- A x0
- B d0
- C x1
- D d1
- E w1

Function Parameters

Which register is used for the function parameter `val3` in:

```
float func(double val1, double* val2, float val3,
float* val4, int val5, double val6)
```

- A s1
- B s2
- C s3
- D w2
- E w1

Function Parameters

Which register is used for the function parameter `val4` in:

```
float func(double val1, double* val2, float val3,
float* val4, int val5, double val6)
```

x1

Function Parameters

Which register is used for the function parameter `val5` in:

```
float func(double val1, double* val2, float val3,
float* val4, int val5, double val6)
```

w2

Function Parameters

Which register is used for the function parameter `val6` in:

```
float func(double val1, double* val2, float val3,
float* val4, int val5, double val6)
```

d2

Instruction Decoding

		.text .global compute compute:
0 0 0 0		cmp x2, x8
		first: nop
		bge done
		nop nop nop nop nop
		bne first
		nop done:
		mov w19, w28
		ret

Instruction Decoding

cmp x2, x8

ADD/SUB/CMP (with optionally shifted register)

31	30	29	28	27	26	25	24	23 22	21	20 … 16	15 … 10	9 … 5	4 … 0
sf	0	S	0	1	0	1	1	shift	0	Rm	imm6	Rn	Rd
	op											Rn	Rm

op is 0 for add, 1 for sub.

shift is shifting mode, imm6 is shift amount.

flags are to be set if S is 1, otherwise flags are not set.

subs xzr, x2, x8

1110 1011 0000 1000 0000 0000 0101 1111

EB08005F

Instruction Decoding

`mov w19, w28`

AND/ANDS/ORR/EOR (with optionally shifted register)

31	30 29	28	27	26	25	24	23 22	21	20 … 16	15 … 10	9 … 5	4 … 0
sf	opc	0	1	0	1	0	shift	0	Rm	imm6	Rn	Rd
	opc							N				

shift is shifting mode, imm6 is shift amount.

opc = 00 for AND, opc = 11 for ANDS, opc = 01 for ORR, opc = 10 for EOR

Rn Rm

eor w19, wzr, w28

MOV (register)

The mov instruction is a pseudo-instruction provided for convenience and does not directly exist. It is actually an alias for the ORR instruction.

0010 1010 0001 1100 0000 0011 1111 0011

2A1C03F3

Instruction Decoding

		.text .global compute compute:
0 0 0 0		cmp x2, x8
		first: nop
0008		bge done
		nop nop nop nop nop
0020		bne first
		nop done:
0028		mov w19, w28
		ret

Instruction Decoding

bge done

B (conditional)

31	30	29	28	27	26	25	24	23 – 5	4	3 – 0
0	1	0	1	0	1	0	0	imm19	0	cond

$8 + imm19 * 4 = 28$

$imm19 = 8$

0101 0100 0000 .... 0001 0000 1010

5400010A

Instruction Decoding

`bne first`

B (conditional)

31	30	29	28	27	26	25	24	23 … 5	4	3 … 0
0	1	0	1	0	1	0	0	imm19	0	cond

$$20 + imm19 * 4 = 4$$
$$imm19 = -7$$

00111
11000+1
= 11001

0101 0100 1111 .... 1111 0010 0001

54FFFF21

iClicker

Which format is the instruction `I = 0x1e57`?

- A Format 0
- B Format 2
- C Format 3
- D Format 4
- E Format 9

Instruction 1

`I = 0x1e57`

- What is the instruction?
- What should be the value of ALUe?
- What is the value of Ae?
- What is the value of Be?
- What is the value of the selector input of M3?
- What is the value of I1 input of M3?

Instruction 1

```
I = 0x1e57
                0001 1110 0101 0111
```

- What is the instruction?
- What should be the value of ALUe?
- What is the value of Ae?
- What is the value of Be?
- What is the value of the selector input of M3?
- What is the value of I1 input of M3?

Instruction 1

```
I = 0x1e57
              00011 1 1 001 010 111
```

- What is the instruction?
- What should be the value of ALUe?
- What is the value of Ae?
- What is the value of Be?
- What is the value of the selector input of M3?
- What is the value of I1 input of M3?

Instruction 1

I = 0x1e57

00011 1 1 001 010 111

- What is the instruction? `sub r7, r2, #1`
- What should be the value of ALUe? 111
- What is the value of Ae? 010
- What is the value of Be? 001
- What is the value of the selector input of M3? 1
- What is the value of I1 input of M3? 001

Instruction 2

`I = 0x19a8`

- What is the instruction?
- What should be the value of ALUe?
- What is the value of Ae?
- What is the value of Be?
- What is the value of the selector input of M3?
- What is the value of I1 input of M3?

Instruction 2

`I = 0x19a8`

`0001 1001 1010 1000`

- What is the instruction?
- What should be the value of ALUe?
- What is the value of Ae?
- What is the value of Be?
- What is the value of the selector input of M3?
- What is the value of I1 input of M3?

Instruction 2

`I = 0x19a8`

`00011 0 0 110 101 000`

- What is the instruction?
- What should be the value of ALUe?
- What is the value of Ae?
- What is the value of Be?
- What is the value of the selector input of M3?
- What is the value of I1 input of M3?

Instruction 2

`I = 0x19a8`

`00011 0 0 110 101 000`

- What is the instruction? `add r0, r5, r6`
- What should be the value of ALUe? `000`
- What is the value of Ae? `101`
- What is the value of Be? `110`
- What is the value of the selector input of M3? `0`
- What is the value of I1 input of M3? `110`

iClicker

Which format is the instruction `I = 0x3521`?

- A Format 0
- B Format 2
- C Format 3
- D Format 4
- E Format 9

Instruction 3

`I = 0x3521`

- What is the instruction?
- What should be the value of ALUe?
- What is the value of Ae?
- What is the value of Be?
- What is the value of Rw?
- What operation should the ALU be configured to perform?

Instruction 3

```
I = 0x3521
            0011 0101 0010 0001
```

- What is the instruction?
- What should be the value of ALUe?
- What is the value of Ae?
- What is the value of Be?
- What is the value of Rw?
- What operation should the ALU be configured to perform?

Instruction 3

```
I = 0x3521
                    001 10 101 00100001
```

- What is the instruction?
- What should be the value of ALUe?
- What is the value of Ae?
- What is the value of Be?
- What is the value of Rw?
- What operation should the ALU be configured to perform?

Instruction 3

I = 0x3521

001 10 101 00100001

- What is the instruction? `add r5, #33`
- What should be the value of ALUe? 101
- What is the value of Ae? 101
- What is the value of Be? Don't care
- What is the value of Rw? 1
- What operation should the ALU be configured to perform? A + B

Instruction 4

`I = 0x2d16`

- What is the instruction?
- What should be the value of ALUe?
- What is the value of Ae?
- What is the value of Be?
- What is the value of Rw?
- What operation should the ALU be configured to perform?

Instruction 4

`I = 0x2d16`

`0010 1101 0001 0110`

- What is the instruction?
- What should be the value of ALUe?
- What is the value of Ae?
- What is the value of Be?
- What is the value of Rw?
- What operation should the ALU be configured to perform?

Instruction 4

`I = 0x2d16`

`001 01 101 00010110`

- What is the instruction?
- What should be the value of ALUe?
- What is the value of Ae?
- What is the value of Be?
- What is the value of Rw?
- What operation should the ALU be configured to perform?

Instruction 4

`I = 0x2d16`

`001 01 101 00010110`

- What is the instruction? `cmp r5, #22`
- What should be the value of ALUe? `101`
- What is the value of Ae? `101`
- What is the value of Be? `Don't care`
- What is the value of Rw? `0`
- What operation should the ALU be configured to perform? `A - B`

iClicker

Which format is the instruction `I = 0x41f4`?

- A Format 0
- B Format 2
- C Format 3
- D Format 4
- E Format 9

Instruction 5

`I = 0x41f4`

- What is the instruction?
- What should be the value of ALUe?
- What is the value of Ae?
- What is the value of Be?
- What is the value of Rw?
- What operation should the ALU be configured to perform?

Instruction 5

```
I = 0x41f4
                0100 0001 1111 0100
```

- What is the instruction?
- What should be the value of ALUe?
- What is the value of Ae?
- What is the value of Be?
- What is the value of Rw?
- What operation should the ALU be configured to perform?

Instruction 5

I = 0x41f4

010000 0111 110 100

- What is the instruction?
- What should be the value of ALUe?
- What is the value of Ae?
- What is the value of Be?
- What is the value of Rw?
- What operation should the ALU be configured to perform?

Instruction 5

I = 0x41f4

010000 0111 110 100

- What is the instruction? ror r4, r6
- What should be the value of ALUe? 100
- What is the value of Ae? 100
- What is the value of Be? 110
- What is the value of Rw? 1
- What operation should the ALU be configured to perform? A ror B

Instruction 6

I = 0x425d

- What is the instruction?
- What should be the value of ALUe?
- What is the value of Ae?
- What is the value of Be?
- What is the value of Rw?
- What operation should the ALU be configured to perform?

Instruction 6

I = 0x425d
0100 0010 0101 1101

- What is the instruction?
- What should be the value of ALUe?
- What is the value of Ae?
- What is the value of Be?
- What is the value of Rw?
- What operation should the ALU be configured to perform?

Instruction 6

```
I = 0x425d
              010000 1001 011 101
```

- What is the instruction?
- What should be the value of ALUe?
- What is the value of Ae?
- What is the value of Be?
- What is the value of Rw?
- What operation should the ALU be configured to perform?

Instruction 6

I = 0x425d

010000 1001 011 101

- What is the instruction? `neg r5, r3`
- What should be the value of ALUe? 101
- What is the value of Ae? 101
- What is the value of Be? 011
- What is the value of Rw? 1
- What operation should the ALU be configured to perform? -B

iClicker

Which format is the instruction `I = 0x761F`?

- A Format 0
- B Format 2
- C Format 3
- D Format 4
- E Format 9

Instruction 7

`I = 0x761F`

- What is the instruction?
- What should be the value of ALUe?
- What should the value of the selector input of M2 be?
- What should the value of the selector input of M3 be?
- What is the value of the I1 input of M3?
- What is the value of Rw?

Instruction 7

I = 0x761F

0111 0110 0001 1111

- What is the instruction?
- What should be the value of ALUe?
- What should the value of the selector input of M2 be?
- What should the value of the selector input of M3 be?
- What is the value of the I1 input of M3?
- What is the value of Rw?

Instruction 7

`I = 0x761F`

`011 1 0 11000 011 111`

- What is the instruction?
- What should be the value of ALUe?
- What should the value of the selector input of M2 be?
- What should the value of the selector input of M3 be?
- What is the value of the I1 input of M3?
- What is the value of Rw?

Instruction 7

I = 0x761F

011 1 0 11000 011 111

- What is the instruction? `str r7, [r3, #96]`
- What should be the value of ALUe? 111
- What should the value of the selector input of M2 be? 1
- What should the value of the selector input of M3 be? 1
- What is the value of the I1 input of M3? 1100000
- What is the value of Rw? 0

Instruction 8

`I = 0x6A56`

- What is the instruction?
- What should be the value of ALUe?
- What should the value of the selector input of M2 be?
- What should the value of the selector input of M3 be?
- What is the value of the I1 input of M3?
- What is the value of Rw?

Instruction 8

```
I = 0x6A56
            0110 1010 0101 0110
```

- What is the instruction?
- What should be the value of ALUe?
- What should the value of the selector input of M2 be?
- What should the value of the selector input of M3 be?
- What is the value of the I1 input of M3?
- What is the value of Rw?

Instruction 8

I = 0x6A56

011 0 1 01001 010 110

- What is the instruction?
- What should be the value of ALUe?
- What should the value of the selector input of M2 be?
- What should the value of the selector input of M3 be?
- What is the value of the I1 input of M3?
- What is the value of Rw?

Instruction 8

I = 0x6A56

011 0 1 01001 010 110

- What is the instruction? `ldr r6, [r2, #36]`
- What should be the value of ALUe? 110
- What should the value of the selector input of M2 be? 1
- What should the value of the selector input of M3 be? 1
- What is the value of the I1 input of M3? 0100100
- What is the value of Rw? 1